

PSet4 Solutions

Problem 1: Matrix Inversion

(a)

Let

$$A = \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix}.$$

For a 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the inverse is

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

provided that $ad - bc \neq 0$.

The determinant of A is

$$\det(A) = (4)(1) - (2)(3) = 4 - 6 = -2.$$

Since

$$\det(A) = -2 \neq 0,$$

the matrix is invertible.

Therefore,

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 1 & -2 \\ -3 & 4 \end{pmatrix}.$$

Thus,

$$\boxed{A^{-1} = \begin{pmatrix} -\frac{1}{2} & 1 \\ \frac{3}{2} & -2 \end{pmatrix}}.$$

We can verify the answer:

$$AA^{-1} = \begin{pmatrix} 4 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & 1 \\ \frac{3}{2} & -2 \end{pmatrix}.$$

Multiplying,

$$\begin{aligned} &= \begin{pmatrix} 4(-\frac{1}{2}) + 2(\frac{3}{2}) & 4(1) + 2(-2) \\ 3(-\frac{1}{2}) + 1(\frac{3}{2}) & 3(1) + 1(-2) \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \end{aligned}$$

Hence the inverse is correct.

(b)

Let

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its determinant is

$$\det(B) = (2)(1) - (1)(1) = 1.$$

Since

$$\det(B) = 1 \neq 0,$$

the matrix is invertible.

Therefore,

$$B^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Thus,

$$\boxed{B^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}}.$$

To verify,

$$BB^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore,

$$= \begin{pmatrix} 2-1 & -2+2 \\ 1-1 & -1+2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Hence,

$$\boxed{BB^{-1} = I}.$$

Problem 2: Dummy Encoding for Categorical Variables

Suppose a categorical variable has n possible categories, and let

$$D_1, D_2, \dots, D_n$$

be the n corresponding dummy variables. For every observation, exactly one of these dummy variables equals 1, while all the others equal 0. Therefore,

$$D_1 + D_2 + \dots + D_n = \mathbf{1},$$

where $\mathbf{1}$ is exactly the column of ones already included in the design matrix X for the intercept.

If all n dummy variables and the intercept are included in X , one column of X is therefore a linear combination of the others. Hence the columns of X are linearly dependent, so X does not have full column rank and

$$X'X$$

is singular. As a result,

$$(X'X)^{-1}$$

does not exist, so the ordinary least squares formula

$$\hat{\mathbf{b}} = (X'X)^{-1}X'\mathbf{y}$$

cannot be computed in this form.

To avoid this problem, we include only $n - 1$ dummy variables and omit one category as the *reference category*. The intercept then represents the expected outcome for the reference category, while the coefficients on the remaining dummy variables represent differences relative to that reference category.

Thus, using $n - 1$ dummy variables avoids perfect multicollinearity, also known as the *dummy variable trap*.

Problem 3: Solve the System of Equations

(a) System #1

The system is

$$\begin{aligned}x + y + 2z &= 2, \\3x - 2y + z &= 1, \\y - z &= 3.\end{aligned}$$

We solve the system by substitution.

From the third equation,

$$y - z = 3,$$

so

$$y = z + 3.$$

Substitute this into the first equation:

$$x + (z + 3) + 2z = 2.$$

Therefore,

$$x + 3z + 3 = 2,$$

so

$$x + 3z = -1.$$

Hence,

$$x = -1 - 3z.$$

Now substitute

$$x = -1 - 3z$$

and

$$y = z + 3$$

into the second equation:

$$3(-1 - 3z) - 2(z + 3) + z = 1.$$

Expanding,

$$-3 - 9z - 2z - 6 + z = 1.$$

Thus,

$$-9 - 10z = 1.$$

Therefore,

$$-10z = 10,$$

so

$$z = -1.$$

Then

$$y = z + 3 = -1 + 3 = 2,$$

and

$$x = -1 - 3(-1) = 2.$$

Therefore, the solution is

$$\boxed{(x, y, z) = (2, 2, -1)}.$$

We can verify:

$$2 + 2 + 2(-1) = 2,$$

$$3(2) - 2(2) + (-1) = 1,$$

and

$$2 - (-1) = 3.$$

Thus all three equations are satisfied.

(b) System #2

The system is

$$y - x = 1,$$

$$z - 2y + x = 1,$$

$$3z - 4x + y = 10.$$

From the first equation,

$$y = x + 1.$$

Substitute this into the second equation:

$$z - 2(x + 1) + x = 1.$$

Thus,

$$z - 2x - 2 + x = 1,$$

so

$$z - x = 3.$$

Therefore,

$$z = x + 3.$$

Now substitute

$$y = x + 1$$

and

$$z = x + 3$$

into the third equation:

$$3(x + 3) - 4x + (x + 1) = 10.$$

Expanding,

$$3x + 9 - 4x + x + 1 = 10.$$

The left-hand side simplifies to

$$10 = 10.$$

This is an identity, so the third equation does not impose any new restriction. Therefore, one variable is free.

Let

$$x = t, \quad t \in \mathbb{R}.$$

Then

$$y = t + 1$$

and

$$z = t + 3.$$

Hence the full set of solutions is

$$\boxed{(x, y, z) = (t, t + 1, t + 3), \quad t \in \mathbb{R}}.$$

Thus, the system has infinitely many solutions.

In matrix form, the coefficient matrix is

$$A = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ -4 & 1 & 3 \end{pmatrix}.$$

Its determinant is

$$\det(A) = 0,$$

so A is not invertible. Therefore, matrix inversion cannot be used to obtain a unique solution, which is consistent with the fact that the system has infinitely many solutions.

Problem 4: Multiplying by 0

For real numbers, if

$$xy = 0,$$

then at least one of x or y must equal zero. However, this property does not hold for matrices.

It is enough to provide a counterexample. Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Clearly,

$$A \neq 0$$

and

$$B \neq 0.$$

However,

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Multiplying the matrices gives

$$\begin{aligned} AB &= \begin{pmatrix} 1(0) + 0(1) & 1(0) + 0(0) \\ 0(0) + 0(1) & 0(0) + 0(0) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

Thus,

$$\boxed{AB = 0}$$

even though

$$\boxed{A \neq 0 \quad \text{and} \quad B \neq 0}.$$

Therefore, the statement

$$AB = 0 \implies A = 0 \text{ or } B = 0$$

is false for matrices.

Problem 5: Determinants

(a)

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Therefore,

$$\det(A) = (2)(1) - (1)(1) = 2 - 1 = \boxed{1}.$$

(b)

Consider

$$B = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix}.$$

Its determinant is

$$\det(B) = (2)(-2) - (1)(-4).$$

Thus,

$$\det(B) = -4 + 4 = \boxed{0}.$$

Notice that the second row is -2 times the first row, so the rows are linearly dependent. This is also why the determinant is zero.

(c)

Consider

$$D = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Expanding along the first row,

$$\det(D) = 2 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}.$$

Compute each 2×2 determinant:

$$\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 0(1) - 1(1) = -1,$$

$$\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0,$$

and

$$\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1.$$

Therefore,

$$\det(D) = 2(-1) - 1(0) + 1(1).$$

Hence,

$$\boxed{\det(D) = -1}.$$

Problem 6: Cofactor Expansion

Consider

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix}.$$

(a)

The minor M_{ij} is the determinant of the matrix obtained by deleting row i and column j .

First,

$$M_{11} = \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = 0 - 2 = -2.$$

$$M_{12} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1 - 1 = 0.$$

$$M_{13} = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} = 2.$$

For the second row,

$$M_{21} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1 - 2 = -1.$$

$$M_{22} = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = 2 - 1 = 1.$$

$$M_{23} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3.$$

For the third row,

$$M_{31} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1.$$

$$M_{32} = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = 1.$$

$$M_{33} = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -1.$$

Therefore, the matrix of minors is

$$M = \begin{pmatrix} -2 & 0 & 2 \\ -1 & 1 & 3 \\ 1 & 1 & -1 \end{pmatrix}.$$

The cofactors are defined by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The sign pattern is

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}.$$

Therefore,

$$C_{11} = -2, \quad C_{12} = 0, \quad C_{13} = 2,$$

$$C_{21} = 1, \quad C_{22} = 1, \quad C_{23} = -3,$$

and

$$C_{31} = 1, \quad C_{32} = -1, \quad C_{33} = -1.$$

Thus, the cofactor matrix is

$$C = \begin{pmatrix} -2 & 0 & 2 \\ 1 & 1 & -3 \\ 1 & -1 & -1 \end{pmatrix}.$$

(b)

Using cofactor expansion along the first row,

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

Thus,

$$\det(A) = 2(-2) + 1(0) + 1(2).$$

Therefore,

$$\boxed{\det(A) = -2}.$$

(c)

Now expand along the second column.

The entries in the second column are

$$a_{12} = 1, \quad a_{22} = 0, \quad a_{32} = 2.$$

The corresponding cofactors are

$$C_{12} = 0, \quad C_{22} = 1, \quad C_{32} = -1.$$

Therefore,

$$\det(A) = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}.$$

Thus,

$$\det(A) = 1(0) + 0(1) + 2(-1) = \boxed{-2}.$$

This agrees with the answer obtained by expanding along the first row.

(d)

Expansion along the second column requires less work because the second column contains a zero. The term involving $a_{22} = 0$ vanishes immediately, so fewer nonzero products need to be calculated than when expanding along the first row.

(e)

The inverse of a matrix can be found using

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A),$$

where

$$\text{adj}(A) = C^T.$$

From part (a),

$$C = \begin{pmatrix} -2 & 0 & 2 \\ 1 & 1 & -3 \\ 1 & -1 & -1 \end{pmatrix}.$$

Therefore,

$$C^T = \begin{pmatrix} -2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & -3 & -1 \end{pmatrix}.$$

Since

$$\det(A) = -2,$$

we have

$$A^{-1} = -\frac{1}{2} \begin{pmatrix} -2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & -3 & -1 \end{pmatrix}.$$

Thus,

$$A^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \\ -1 & \frac{3}{2} & \frac{1}{2} \end{pmatrix}.$$

Problem 7: Eigenvalues and Eigenvectors

Consider again

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix}.$$

(a)

The trace of a matrix is the sum of its diagonal entries. Therefore,

$$\text{tr}(A) = 2 + 0 + 1.$$

Thus,

$$\boxed{\text{tr}(A) = 3}.$$

(b)

The eigenvalues satisfy the characteristic equation

$$\det(\lambda I - A) = 0.$$

We have

$$\lambda I - A = \begin{pmatrix} \lambda - 2 & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -2 & \lambda - 1 \end{pmatrix}.$$

Therefore,

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 2 & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -2 & \lambda - 1 \end{vmatrix}.$$

Expanding and simplifying gives

$$p(\lambda) = \lambda^3 - 3\lambda^2 - 2\lambda + 2.$$

To verify that $\lambda = -1$ is a root,

$$p(-1) = (-1)^3 - 3(-1)^2 - 2(-1) + 2.$$

Thus,

$$p(-1) = -1 - 3 + 2 + 2 = 0.$$

Hence,

$$\lambda = -1$$

is indeed an eigenvalue.

Factor out $(\lambda + 1)$:

$$\lambda^3 - 3\lambda^2 - 2\lambda + 2 = (\lambda + 1)(\lambda^2 - 4\lambda + 2).$$

Thus,

$$(\lambda + 1)(\lambda^2 - 4\lambda + 2) = 0.$$

One eigenvalue is

$$\boxed{\lambda_1 = -1}.$$

For the remaining two eigenvalues, solve

$$\lambda^2 - 4\lambda + 2 = 0.$$

Using the quadratic formula,

$$\lambda = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(2)}}{2}.$$

Therefore,

$$\lambda = \frac{4 \pm \sqrt{16 - 8}}{2} = \frac{4 \pm \sqrt{8}}{2}.$$

Since

$$\sqrt{8} = 2\sqrt{2},$$

we obtain

$$\lambda = 2 \pm \sqrt{2}.$$

Hence, the three eigenvalues are

$$\boxed{\lambda_1 = -1, \quad \lambda_2 = 2 - \sqrt{2}, \quad \lambda_3 = 2 + \sqrt{2}}.$$

As a check, their sum is

$$-1 + (2 - \sqrt{2}) + (2 + \sqrt{2}) = 3,$$

which agrees with

$$\text{tr}(A) = 3.$$

(c)

For $\lambda = -1$, an eigenvector $\mathbf{v}$ must satisfy

$$A\mathbf{v} = -\mathbf{v}.$$

Equivalently,

$$(A + I)\mathbf{v} = 0.$$

We have

$$A + I = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \end{pmatrix}.$$

Let

$$\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Then

$$\begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

This gives

$$3x + y + z = 0,$$

$$x + y + z = 0,$$

and

$$x + 2y + 2z = 0.$$

Subtracting the second equation from the first gives

$$2x = 0,$$

so

$$x = 0.$$

Then

$$y + z = 0,$$

so

$$y = -z.$$

Choose $z = 1$. Then

$$y = -1.$$

Therefore, one eigenvector is

$$\boxed{\mathbf{v} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}}.$$

Now verify that $A\mathbf{v} = -\mathbf{v}$:

$$A\mathbf{v} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Thus,

$$A\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

On the other hand,

$$-\mathbf{v} = -\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

Therefore,

$$\boxed{A\mathbf{v} = -\mathbf{v}},$$

confirming that $\mathbf{v}$ is an eigenvector corresponding to $\lambda = -1$.

(d)

According to the criterion in the problem, a matrix is positive definite when all of its eigenvalues are positive.

The eigenvalues of A are

$$-1, \quad 2 - \sqrt{2}, \quad 2 + \sqrt{2}.$$

Although

$$2 - \sqrt{2} > 0$$

and

$$2 + \sqrt{2} > 0,$$

we have

$$-1 < 0.$$

Therefore, not all eigenvalues are positive, so

$$\boxed{A \text{ is not positive definite.}}$$